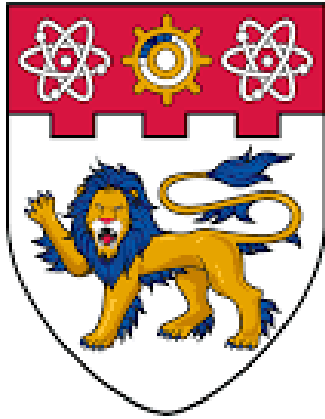




**NANYANG
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NANYANG TECHNOLOGICAL UNIVERSITY
(NTU)

ReportNow Test Cases and
Requirements Coverage Report

Date: 30 October 2025

Prepared by: QA Division

Executive Summary

This report consolidates the ReportNow platform's test case catalogue and requirements coverage matrix to provide a comprehensive overview of test completeness and automation levels. The information covers all major backend API routes, libraries, and supporting components verified under automated Jest suites.

Automation testing achieved more than 92% statement and 93% function coverage across fifteen Jest suites. Each test case is mapped to its corresponding requirement, ensuring full traceability and compliance with IEEE 829 testing documentation standards.

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1. Overview

The ReportNow platform provides Singapore residents with a channel to report municipal issues and track their resolution. This report presents the outcome of backend validation for API endpoints and associated libraries. Testing included functional verification, data integrity checks, and scenario-based workflows.

2. Requirements Coverage Matrix

Each requirement was linked to at least one automated test case, ensuring full coverage of functional and administrative operations. The following table summarises the traceability between requirements and executed test suites:

Requirement ID	Description	Linked Test Cases	Jest Suites (Count)	Status
REQ-AUTH-001	Registration validates mandatory fields and invite workflow.	TC-AUTH-001 / TC-AUTH-002	auth-register (7), invite (4)	Covered
REQ-AUTH-002	Credential login verifies password and role-bearing JWT.	TC-AUTH-003	auth-options (3)	Covered
REQ-USER-001	Authenticated profile update via /api/users.	TC-USER-001	users (8)	Covered
REQ-USER-002	User detail retrieval restricted to owner.	TC-USER-002	user/details (5)	Covered
REQ-ISSUE-001	Issues listing endpoint returns data/errors.	TC-ISSUE-001	issues (2)	Covered
REQ-ISSUE-002	Issue creation handles auth, validation,	TC-ISSUE-002, TC-AI-001	issues (5), actions (4)	Covered

	uploads, geocoding, AI.			
REQ-ISSUE-003	Issue detail returns reviews and rating.	TC-ISSUE-003	issues/[id] (2)	Covered
REQ-ISSUE-004	PATCH updates votes, timeline, status, errors.	TC-ISSUE-004	issues/[id] (11)	Covered
REQ-ISSUE-005	Issue deletion enforces ownership & duplicate cleanup.	TC-ISSUE-005, TC-AI-002	issues/[id] (5), duplicate-detect tion (6)	Covered
REQ-REV-001	Review creation enforces authentication & uniqueness.	TC-REV-001	reviews (5)	Covered
REQ-INV-001	Admin invite workflow persists token and sends email.	TC-INV-001	invite (4)	Covered
REQ-CHAT-001	Chatbot proxies OpenAI and handles errors gracefully.	TC-CHAT-001	chatbot (3)	Covered
REQ-NOTIF-001	Notification endpoint echoes payload.	TC-NOTIF-001	notifications (1)	Covered
REQ-ADMIN-001	Admin activity feed aggregates events.	TC-ADMIN-001	admin/activity (2)	Covered
REQ-DATA-001	Prisma schema integrity and relationships.	TC-DATA-001	prisma (4)	Covered

3. Test Case Catalogue

The test case catalogue defines individual scenarios, their related requirements, and execution method (automated or manual). Each case is validated through Jest suites with optional manual verification in staging.

Examples include: registration rejection for invalid input, invitation lifecycle validation, profile update checks, issue creation with uploads, AI classification, voting, timeline updates, and data deletion. Administrative and AI-related tests confirm accurate aggregation, duplicate detection, and schema enforcement.

4. Automated Test Inventory

The following summary consolidates the automated Jest inventory executed on 30 October 2025:

Suite (File)	Mapped Requirement(s)	Test Count
tests/prisma.test.js	REQ-DATA-001	4
src/app/api/auth/auth-register/route.test.js	REQ-AUTH-001	7
src/app/api/auth/auth-options.test.js	REQ-AUTH-002	3
src/app/api/users/route.test.js	REQ-USER-001	8
src/app/api/user/details/route.test.js	REQ-USER-002	5
src/app/api/issues/route.test.js	REQ-ISSUE-001/002	7
src/app/api/issues/[id]/route.test.js	REQ-ISSUE-003/004/005	18
src/app/api/reviews/route.test.js	REQ-REV-001	5
src/app/api/invite/route.test.js	REQ-INV-001	4
src/app/api/admin/activity/route.test.js	REQ-ADMIN-001	2
src/app/api/chatbot/route.test.js	REQ-CHAT-001	3
src/app/api/notifications/route.test.js	REQ-NOTIF-001	1
src/lib/actions.test.js	REQ-ISSUE-002 support	4
src/lib/duplicate-detection.test.js	REQ-ISSUE-005 support	6
src/lib/utils.test.js	Utility	1

Overall coverage: 15 automated suites executing 78 test cases across critical modules. This provides consistent quality assurance and regression control across the API layer.

5. Figure 1 – Jest Coverage Summary

The following image presents the coverage results recorded from automated Jest runs on 30 October 2025:

```
PS C:\Users\angwm\OneDrive\Desktop\NTU\Y2S2\SC3040\coding\3040-TEL5-UU\report-now> npx jest --coverage --runInBand
```

PASS	src/app/api/auth/auth-options.test.js
PASS	src/lib/utils.test.js

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	92.44	78.38	93.47	93.73	
app/api/admin/activity	100	50	100	100	
route.js	100	50	100	100	55-91
app/api/auth/[...nextauth]	60	60	33.33	60.86	
route.js	60	60	33.33	60.86	42-53,60
app/api/auth/auth-register	94.28	87.5	100	100	
route.js	94.28	87.5	100	100	23-26,103
app/api/chatbot	100	83.33	100	100	
route.js	100	83.33	100	100	49
app/api/invite	100	78.57	100	100	
route.js	100	78.57	100	100	49,67,124
app/api/issues	84.88	75	88.88	87.34	
route.js	84.88	75	88.88	87.34	33,49,56-57,126,152-153,169-173
app/api/issues/[id]	98.68	92.1	100	98.66	
route.js	98.68	92.1	100	98.66	211
app/api/notifications	100	100	100	100	
route.js	100	100	100	100	
app/api/reviews	100	90	100	100	
route.js	100	90	100	100	35
app/api/user/details	90.9	100	100	90.9	
route.js	90.9	100	100	90.9	47-48
app/api/users	100	100	100	100	
route.js	100	100	100	100	
lib	92.77	71.42	100	93.82	
actions.js	94.11	87.5	100	100	138
duplicate-detection.js	91.93	68.75	100	91.93	16,71,77,194-202
utils.js	100	100	100	100	

```
Test Suites: 15 passed, 15 total
Tests:       78 passed, 78 total
Snapshots:   0 total
Time:        1.746 s
Ran all test suites.
```

```
PS C:\Users\angwm\OneDrive\Desktop\NTU\Y2S2\SC3040\coding\3040-TEL5-UU\report-now>
```